

task-1

RAM Reseating Practical – Step-by-Step

Safety: Shut down the computer completely, unplug the power adapter/cable, and if it is a laptop, disconnect the battery if it is safely removable. Avoid touching the gold contacts on the RAM.

Procedure:

1. Shut down the PC/laptop completely.
2. Disconnect the power cable/charger and all external devices.
3. Open the computer case/back cover using the appropriate screwdriver.
4. Locate the **RAM (memory) slots** on the motherboard.
5. Note the position and orientation of the RAM module before removing it.
6. Push the **retaining clips** on both sides of the RAM slot outward.
7. Carefully remove the RAM module by holding it from its edges.
8. Inspect the RAM and slot for **dust, physical damage, or bent/dirty contacts**.
9. Align the notch on the RAM module with the key in the RAM slot.
10. Press the RAM module firmly and evenly into the slot until the retaining clips lock into place.
11. Close the computer case/back cover.
12. Reconnect the power and start the computer.
13. Check whether the PC boots normally and verify the installed RAM in **BIOS/UEFI or Windows Task Manager → Performance → Memory**.

Observation Table

Item	Observation
RAM slots located	Yes
RAM module removed	Yes
RAM module reseated	Yes
Dust/damage observed	No major damage observed
RAM clips locked correctly	Yes

Item	Observation
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PC booted after reseating	Yes
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Any error/beep observed	None
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Result:

The RAM module was successfully removed and re-inserted. After reseating, the computer was checked for proper booting and memory detection. Reseating RAM can help resolve problems caused by a loose or improperly connected memory module.

Task -2

GPU Troubleshooting Checklist – Gaming PC

Step	Troubleshooting Action	What to Check
1	☒ Identify the symptoms	Check for no display, screen flickering, artifacts, crashes, low FPS, or GPU-related error messages.
2	☒ Check the display connection	Make sure the HDMI/DisplayPort cable is connected to the GPU's video port , not the motherboard port. Test another cable or monitor if available.
3	☒ Inspect physical connections	Turn off and unplug the PC. Check that the GPU is firmly seated in the PCIe slot and that all required PCIe power connectors are securely connected.
4	☒ Check GPU drivers	In Windows Device Manager → Display adapters , check for errors. Update or reinstall the correct GPU driver from the manufacturer's official source.
5	☒ Monitor GPU temperature	Run a game or GPU stress test and check temperatures. Excessive temperatures may indicate poor airflow, blocked fans, or dust buildup.
6	☒ Check power supply	Verify that the PSU has sufficient wattage and that the GPU receives the required power. Unexpected shutdowns or crashes can indicate power problems.
7	☒ Test the GPU	Run a graphics-intensive game or benchmark. Look for artifacts, crashes, freezes, or abnormal performance. If possible, test the GPU in another compatible PC.

Step	Troubleshooting Action	What to Check
 8	Confirm the fix	Restart the PC, verify that the GPU is detected correctly, and run the same game/test again. Confirm stable display, normal temperatures, and no crashes or artifacts.

Task-3

Storage Issue Simulation – SATA/NVMe Drive

Safety: Perform this only on a test PC. Shut the system down completely and disconnect power before physically disconnecting a SATA drive. For an NVMe drive, remove power and follow the motherboard/laptop manufacturer's procedure before removing the module.

Procedure

1. Shut down the test computer completely.
2. Disconnect the power cable.
3. Open the computer case and locate the **SATA SSD/HDD** or **NVMe SSD**.
4. Before disconnecting it, enter BIOS/UEFI and check **Storage/SATA/NVMe Information**. Record the drive name/model.
5. Power off again and disconnect the drive:
 - **SATA:** Disconnect the SATA data cable or power cable.
 - **NVMe:** Remove the retaining screw and carefully remove the SSD.
6. Power on the system without the drive.
7. Check the BIOS/UEFI storage section and observe whether the drive is missing.
8. If the disconnected drive contained the operating system, a typical message may be **“No bootable device,” “Boot device not found,”** or **“Operating system not found.”**
9. Shut down the system and reconnect/reinstall the drive securely.
10. Start the computer and enter BIOS/UEFI again.
11. Verify that the drive is detected with its **model name and capacity**.
12. Save the settings if required and boot into Windows.
13. Confirm that the drive is accessible in **File Explorer/Disk Management**.

Observation Table

Condition	BIOS/Screen Observation
Before disconnecting	SSD/HDD detected normally
Drive disconnected	Drive not listed in BIOS/UEFI
Boot attempt without drive	“No bootable device” / “Boot device not found”
Drive reconnected	SSD/HDD detected again
After successful boot	Operating system loads normally and drive is accessible

Task-4

Random PC Restart – Component to Check First

Component to Check First: RAM

I would check the **RAM first** because faulty, loose, or incompatible memory can cause system instability, crashes, and unexpected restarts. RAM is also relatively easy to inspect and reseal before moving on to GPU or storage diagnostics.

Diagnostic Steps for RAM

1. Check the symptoms

- Note when the restart occurs—during gaming, heavy applications, or even when idle.
- Look for BSODs, freezing, or beep codes during startup.

2. Power off and inspect the RAM

- Shut down the PC and disconnect the power.
- Open the case and check that the RAM modules are properly seated.
- Look for visible damage or excessive dust.

3. Reseat the RAM

- Remove the RAM module carefully.
- Reinsert it firmly until the retaining clips lock into position.
- If there are multiple modules, test them **one at a time**.

4. Run a memory diagnostic

- Use **Windows Memory Diagnostic**: press Win + R, type mdsched.exe, and restart to perform the test.
- For more thorough testing, a bootable memory-testing utility such as MemTest86 can be used.

5. Check BIOS/UEFI

- Enter BIOS/UEFI and verify that the correct **RAM capacity and speed** are detected.
- If an overclock/XMP/EXPO profile is enabled, temporarily return memory settings to default.

6. Confirm the fix

- Boot Windows and run the applications that previously caused the restart.
- Monitor the system for stability and confirm that no unexpected restarts occur.

If RAM Tests Good

Then I would move on to the **GPU** (drivers, seating, temperature, power) and then **storage** (SMART health, cables, and disk errors). A random restart can also be caused by the **PSU or CPU overheating**, so those should be considered if RAM/GPU/storage checks do not identify the problem.

Conclusion: RAM → GPU → Storage is a practical diagnostic order for these three components, starting with the component that is relatively easy to test and can cause broad system instability.